

LOGAN JAMES MARSHALL

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EDUCATION

University of Colorado Boulder, **Boulder, Colorado**
Bachelor's--Accelerated Master's Program
Master of Science, Mechanical Engineering **Expected May 2028**
Bachelor of Science, Mechanical Engineering, Computer Science Minor **Expected May 2027**
• GPA: 3.99 / 4.00

Relevant Coursework: Component Design, Dynamics, Fluid Mechanics, Heat Transfer, Computational Methods, Modeling Human Movement, Circuits and Electronics, Algorithms, Artificial Intelligence, Mechatronics

Awards & Scholarships: J. Perry Bartlett STEM Scholarship, CU Esteemed Scholars – Horace M. Hale Scholarship

TECHNICAL SKILLS

Mechanical Design: SolidWorks (CSWA, CSWP certified), Onshape, GD&T, technical drawings, AutoCAD

Electronics: Arduino, Raspberry Pi, DC Motors and Servos, Soldering, Circuit Design

Fabrication: 3D Printing, CNC Machining, Milling, Lathes, MIG Welding, Manufacturing Processes

Programming: MATLAB, C++, Python, Excel, Microsoft Office

ENGINEERING PROJECTS

Non-Contact Position Sensing for Aerospace Applications - Lead Electromechanical Engineer **Aug 2026 - Present**
Senior Design Capstone sponsored by Los Alamos National Laboratory

- Leading electromechanical design of a non-contact sensing system measuring angular position and 3D displacement of a rotating object through a non-ferrous barrier at 1 in. standoff
- Evaluating candidate sensing technologies against 0.010 in. resolution, 200 in./sec data acquisition rate, and high-dynamic aerospace environment requirements
- Owning mechanical integration CAD and laboratory test setup, working with LANL R&D mentors through milestone design reviews toward a functioning prototype

Autonomous Soccer Robot, Mechatronics - Project Lead **Aug 2026 - Present**

- Designing an autonomous robot utilizing Arduino, Raspberry Pi, and additional microprocessors; integrated onboard OpenCV vision to locate a ball and goals for head-to-head competition in 4-minute matches within a 14 x 12 x 12 in. envelope and \$200 budget
- Prototyping to a fixed milestone schedule ending in a competition runoff and full technical report

Drill Powered Vehicle, Component Design - Systems Engineer **Jan 2025 - May 2025**

- Designed 10+ CAD components and the final SolidWorks assembly; performed FEA and hand calculations to verify factor of safety and gear ratio
- Fabricated and welded custom parts into a final vehicle under 50 lbs, managing a \$200 budget across a 6-person team and presenting PDR and CDR to shop professionals
- **Placed 4th** in the endurance competition, traveling 2.25 miles in 30 minutes on drill batteries

EXPERIENCE

Pure Fishing - AI Development Engineering Intern **May 2026 - Aug 2026 , May 2025 - Aug 2025**

- Integrated a custom Bland AI voice agent and Gorgias AI chat and email bot into customer service CRM platform
- Raised the automated ticket resolution rate to **29%**, removing repetitive requests from the human queue
- **Won 1st place out of five teams** in the internship program; presented AI initiatives at the Charleston, SC headquarters town hall to the CEO, CFO, CPO, and 300+ employees
- Built an AI-powered Retrieval-Augmented Generation (RAG) system in Python, integrating external APIs and applying LoRA-based fine-tuning
- Led evaluation, prototyping, and deployment of an end-to-end system in 12 weeks, increasing storyboard and script production **450%** at under **\$60/month**

C-CONNEX - Junior Design Engineer **May 2023 - Dec 2025**

- Designed 10+ fiber optic hardware assemblies in SolidWorks, including racks, panels, and adapter plates using technical drawings and GD&T to support manufacturable designs.
- Improved manufacturability and durability of designs through rapid 3D-printed prototyping, accelerating client approval cycles

LEADERSHIP & ACTIVITIES

CU Boulder - Fluid Mechanics, Course Assistant **Aug 2026 - Present**

- Grade weekly homework and exams for 35 students; attend lecture to facilitate discussion and help students work through complex problems

Design Build Fly (DBF) Competition Team - Design Team Member **Aug 2024 - Dec 2025**

- Modeled RC airplane components in SolidWorks and conducted whiffletree analysis for distributed wing load testing; CU placed in the top half of 92 teams at the AIAA international competition

Tau Beta Pi Engineering Honor Society - Membership Director **Aug 2024 - Dec 2025**

- Increased active membership by **125%** through targeted outreach, recruitment events, and participation analytics